

9 The harlequin ladybird, *Harmonia axyridis*, is an insect that shows discontinuous variation in colour pattern and continuous variation in body size.

- (a) Within a species, the variation that is observed for a particular characteristic can be described as discontinuous or continuous.
Discontinuous variation has a different genetic basis from continuous variation.

State **two** differences between discontinuous variation and continuous variation, **other than** having a different genetic basis.

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- (b) Fig. 9.1 shows three of the colour patterns seen in *H. axyridis* and the percentage of each in a population.

colour pattern	 <i>conspicua</i>	 <i>spectabilis</i>	 <i>succinea</i>
percentage	5	13	82

Fig. 9.1

Suggest the genetic basis of the variation in colour pattern in *H. axyridis*.

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- (c) The body size of adult *H. axyridis* varies from 5 mm to 8 mm. Adult ladybirds do **not** grow or change size. Adults develop from larvae that hatch from eggs.

State **two** environmental factors that may affect adult body size in *H. axyridis*.

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